Bayesian analysis of field goal attempts in American football

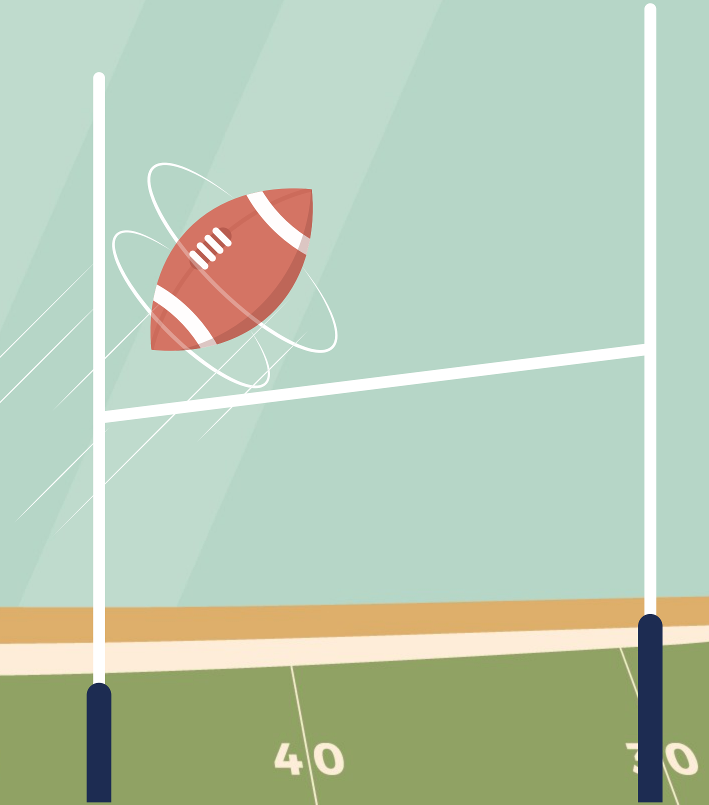
Introduction/ Motivation

Field goals are a critical part of NFL strategy

Coaches are faced with the decision: Attempt a FG or go for it on 4th down in order to win a game

Kickers are kicking father than ever, but many teams such as the Lions and Commanders opt to go for it on 4th down instead of kicking a FG

While other teams, like the Steelers, win games by exclusively scoring with FGs

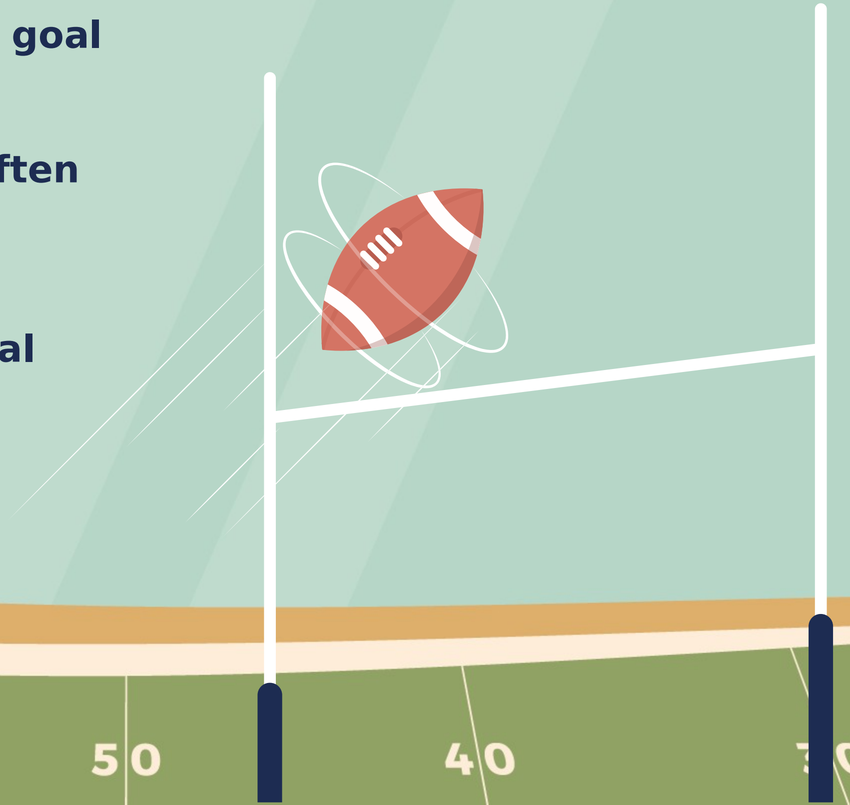


Introduction/ Motivation

Obviously, distance affects field goal success, but by how much?

What distance do teams most often take a field goal from?

To answer these, we will apply Bayesian Inference and Logistical Regression



Methods

- **Data Source:**
  - Aggregated NFL field goal data from the 2024 season
  - Categorized into five distance bins:
    0–19.5, 19.5-29.5, 29.5-39.5, 39.5–49.5, and 49.5-100 yards
  - Each bin includes number of successful kicks and attempts
- **Statistical Approach:**
  - Beta Distribution using Bayesian Inference and Laplace approximation
  - Uninformative prior
  - Posterior: MAP estimate (Maximum A Posteriori)
- **Visualization**:
  - PDF and CDF of attempt distances modeled by a Beta distribution
  - Logistic Regression Fit (Success Probability vs Distance)

Initial Data

Table 1: Field Goal Makes (FGM) and Attempts (FGA) for the 2024 season

Distance Bin (yds)	Makes (FGM)	Attempts (FGA)	Success Rate
0-19.5	2	2	100.00%
19.5-29.5	229	236	97.03%
29.5-39.5	280	297	94.28%
39.5-49.5	231	301	76.74%
49.5-100	195	279	69.89%

Modeling Field Goal Attempts & Success

1. **Normalize Distances**

- Field goal distances are modeled as “X” which is unitless
- X = (distance / 100 yards) so that $X \in (0,1)$
- Assume X follows beta distribution with unknown parameters α and *β*.
- $X \sim \text{Beta}(\alpha,\beta)$
- **2. Bin the Data**
- Exact distance is not provided; we know how many kicks fall within 5 distance bins
- For each bin, the proportion of kicks φ_k comes from integrating Beta density over that bin

Modeling Field Goal Attempts & Success

1. Equation: $\phi_k = \int_{a_k}^{b_k} f_{\text{Beta}}(x|\alpha,\beta)\,dx = F_{\text{Beta}}(b_k|\alpha,\beta) - F_{\text{Beta}}(a_k|\alpha,\beta)$

ϕ_k = Probability (X falls in bin k)

Simplified:

$\phi_k = F_{\text{Beta}}(b_k|\alpha,\beta) - F_{\text{Beta}}(a_k|\alpha,\beta)$

(F_{Beta} is the Beta CDF)

Modeling Field Goal Attempts & Success

3. Modeling Attempt Counts

- The number of **attempts** in each bin follows a **Multinomial** distribution based on the proportions φ_k.

4. Modeling Successes

- Given attempts, the number of **made** kicks in each bin follows a **Binomial** distribution.
- The probability of making a field goal depends on distance through a logistic function with parameters γ_0 and γ_1.

Modeling Field Goal Attempts & Success

Attempt Count Model:

$$n = (n_1, \ldots, n_K) \sim Multinomial(N, \phi)$$

N = total number of attempts

$\phi = (\phi_1, \ldots, \phi_K)$.

Successes Model:

$y_k \sim \mathrm{Binomial}(n_k, p_k)$

p_k = probability of success for bin k, and n_k = number of attempts in bin k .

Modeling Field Goal Attempts & Success

1. Logistic Function Equation:

$$p(x|\gamma_0,\gamma_1) = \frac{1}{1+e^{-(\gamma_0+\gamma_1 x)}}$$

Successes Model:

$$p_k = \frac{1}{\phi_k}\int_{a_k}^{b_k} p(x|\gamma_0,\gamma_1)\, f_{\mathrm{Beta}}(x|\alpha,\beta)\, dx$$

Modeling Field Goal Attempts & Success

5. Inference Goal

- Infer the 4 unknown parameters:

γ_0, γ_1 (make probability curve)

α , β (attempt distance distribution)

6. Bayesian Appraoch

- Use Bayes' theorem to combine data and a prior.
- Since this model is **nonlinear**, use **Laplace's approximation** around the posterior mode to estimate uncertainty. (Find MAP estimate using **fminunc**)

Modeling Field Goal Attempts & Success

Bayesian Inference: posterior $\propto$ likelihood $\times$ prior

$$f_{\gamma_0,\gamma_1,\alpha,\beta|m,n}(\gamma_0,\gamma_1,\alpha,\beta|m,n) \propto P(m,n|\gamma_0,\gamma_1,\alpha,\beta)\, f_{\gamma_0,\gamma_1,\alpha,\beta}(\gamma_0,\gamma_1,\alpha,\beta)$$

Likelihood: $P(m,n|\gamma_0,\gamma_1,\alpha,\beta) = P(m|n,\gamma_0,\gamma_1,\alpha,\beta)\, P(n|\gamma_0,\gamma_1,\alpha,\beta)$

Prior: $f_{\gamma_0,\gamma_1,\alpha,\beta}(\gamma_0,\gamma_1,\alpha,\beta) \propto \frac{1}{\alpha\beta}$

since $\gamma_0 \in (-\infty,\infty)$, $\gamma_1 \in (-\infty,\infty)$, $\alpha \in (0,\infty)$, and $\beta \in (0,\infty)$

Modeling Field Goal Attempts & Success

1. Likelihood: $P(m, n|\gamma_0, \gamma_1, \alpha, \beta) = P(m|n, \gamma_0, \gamma_1, \alpha, \beta)\, P(n|\gamma_0, \gamma_1, \alpha, \beta)$

$$P(m|n, \gamma_0, \gamma_1, \alpha, \beta) = P(m|n, \gamma_0, \gamma_1) = \prod_{k=1}^{K} P_{\mathrm{Bin}}(m_k|n_k, p_k)$$

$$P(n|\gamma_0, \gamma_1, \alpha, \beta) = P(n|\alpha, \beta) = P_{\mathrm{MN}}(n|\phi)$$

P_{Bin} denotes a binomial PMF

P_{MN} denotes a multinomial PMF

Modeling Field Goal Attempts & Success

1. Laplace Approximation Step (Log-Posterior to find Parameters):

$$\ln f_{\gamma_0,\gamma_1,\ln\alpha,\ln\beta|m,n}(\gamma_0,\gamma_1,\ln\alpha,\ln\beta\,|m,n)$$

$$\propto \ln P(m,n|\gamma_0,\gamma_1,\ln\alpha,\ln\beta) + \ln f_{\gamma_0,\gamma_1,\ln\alpha,\ln\beta}(\gamma_0,\gamma_1,\ln\alpha,\ln\beta)$$

Log-Likelihood:

$$\ln P(m,n|\gamma_0,\gamma_1,\ln\alpha,\ln\beta) = \sum_{k=1}^{K}\ln P_{\text{Bin}}(m_k|n_k,p_k) + \ln P_{\text{MN}}(n|\phi)$$

$$\ln f_{\gamma_0,\gamma_1,\ln\alpha,\ln\beta|m,n}(\gamma_0,\gamma_1,\ln\alpha,\ln\beta\,|m,n) \propto \sum_{k=1}^{K}\ln P_{\text{Bin}}(m_k|n_k,p_k) + \ln P_{\text{MN}}(n|\phi)$$

30 40 50 40 30

Important Pseudo-Code

```
% Data setup
interval_boundaries = [0, 19.5, 29.5, 39.5, 49.5, 100]; % In yards
a = interval_boundaries(1:end-1)/100; % Lower bounds (normalized)
b = interval_boundaries(2:end)/100;   % Upper bounds (normalized)
n = [2, 236, 297, 301, 279];         % FGA data
m = [2, 229, 280, 231, 195];         % FGM data

% Run the Bayesian estimation
theta_0 = [logit(mean(m./n)); -0.1; log(2); log(2)];
options = optimoptions('fminunc', 'Display', 'iter', 'Algorithm', 'quasi-newton');
[theta_MAP, ~, ~, ~, ~, negH] = fminunc(@(theta) neg_log_post(theta, a, b, n, m), theta_0, options);
alpha_est = exp(theta_MAP(3));
beta_est = exp(theta_MAP(4));
```

Finds MAP estimates of parameters

Minimizes the negative log-posterior

Important Pseudo-

```
%% Helper functions
function neg_log_post_pdf_val = neg_log_post(theta, a, b, n, m)
    % Unpack and transform parameters
    gamma_0 = theta(1);
    gamma_1 = theta(2);
    ln_alpha = theta(3);
    ln_beta  = theta(4);

    alpha = exp(ln_alpha);
    beta  = exp(ln_beta);

    % Fine grid for integration
    dx = 1e-3;
    xgrid = 0:dx:1;

    % Precompute beta PDF and logistic function
    f_X = betapdf(xgrid, alpha, beta);
    p = 1./(1 + exp(-(gamma_0 + gamma_1*xgrid)));
```

$$p_k = \frac{1}{\phi_k}\int_{a_k}^{b_k} p(x|\gamma_0,\gamma_1)\, f_{\text{Beta}}(x|\alpha,\beta)\, dx$$

```
    % Number of intervals
    K = numel(a);

    % Proportion of X values in each interval
    phi = betacdf(b, alpha, beta) - betacdf(a, alpha, beta);

    % Success probability within each interval
    p_k = nan(1, K);
    for k=1:K
        idx = xgrid >= a(k) & xgrid <= b(k);
        p_k(k) = trapz(xgrid(idx), p(idx).*f_X(idx))/phi(k);
    end

    % Negative log-posterior
    neg_log_post_pdf_val = -sum(log(binopdf(m, n, p_k))) - log(mnpdf(n, phi));
end
```

$$\ln f_{\gamma_0,\gamma_1,\ln\alpha,\ln\beta|m,n}(\gamma_0,\gamma_1,\ln\alpha,\ln\beta\,|m,n) \propto \sum_{k=1}^{K} \ln P_{\text{Bin}}(m_k|n_k,p_k) + \ln P_{\text{MN}}(n|\phi)$$

Results

Table 2: Probability of taking a kick in each bin

Probability of taking a kick in each bin:
Bin 1 (0.0-19.5 yds): 0.0263
Bin 2 (19.5-29.5 yds): 0.1477
Bin 3 (29.5-39.5 yds): 0.2951
Bin 4 (39.5-49.5 yds): 0.2988
Bin 5 (49.5-100.0 yds): 0.2322

$$\phi_k = F_{Beta}\,(b_k|\alpha,\beta) - F_{Beta}(a_k|\alpha,\beta)$$

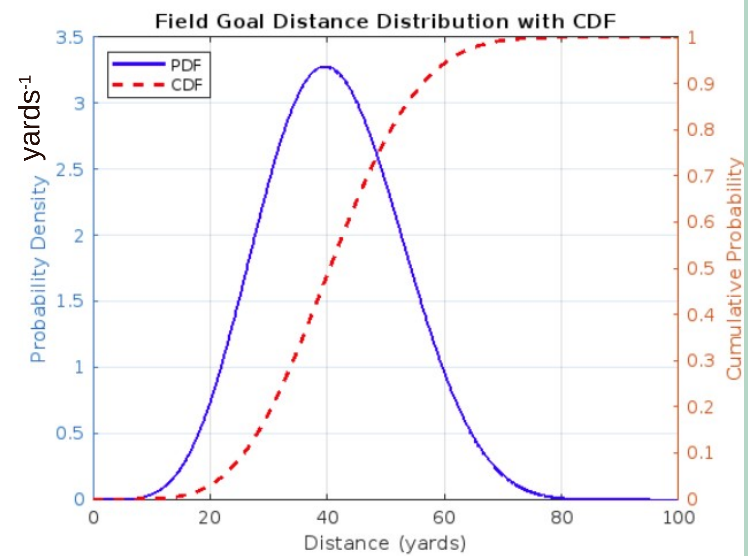


Figure 1: PDF and CDF of Field Goal Distance

Results

Table 3: Shape Parameters

```
=== Beta Distribution Parameters ===
Alpha (α) = 6.8163
Beta (β) = 9.8827
```

Table 4: Distribution FG Statistics

```
=== Distribution Statistics ===
Mean distance: 40.8 yards
Mode distance: 39.6 yards
Standard deviation: 11.7 yards
```

α: tendency toward shorter distances

β: tendency toward longer distances

$\frac{\alpha}{\alpha+\beta}$: mean distance (average fg attempt)

$\frac{\alpha-1}{\alpha+\beta-2}$: mode (most common attempt)

$\sqrt{\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}}$: Standard deviation

Everything * 100 to convert to yards

Results

Table 5: γ_0 and γ_1 coefficient outputs

Logistic regression coefficients using expected yardage:

	Estimate	SE	tStat	pValue
(Intercept)	5.1404	0.40784	12.604	2.0043e-36
Distance	-0.078986	0.0084759	-9.3188	1.1762e-20

Table 6: Success Rate

Distance (yds)	SuccessRate (%)
0	99.4
20	97.2
30	94.1
40	87.9
50	76.7
60	59.9

$$p(x|\gamma_0,\gamma_1) = \frac{1}{1+e^{-(\gamma_0+\gamma_1 x)}}$$

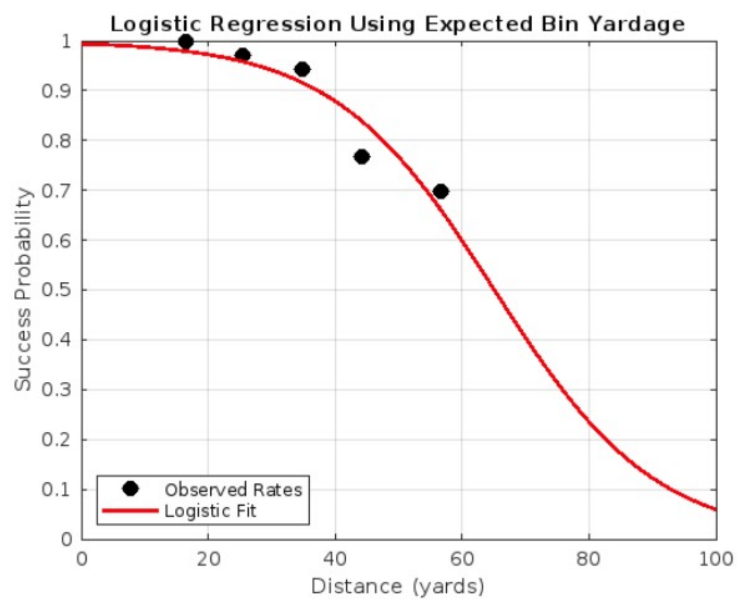


Figure 2: Logistic Regression using Expected Bin Yardage

Discussion

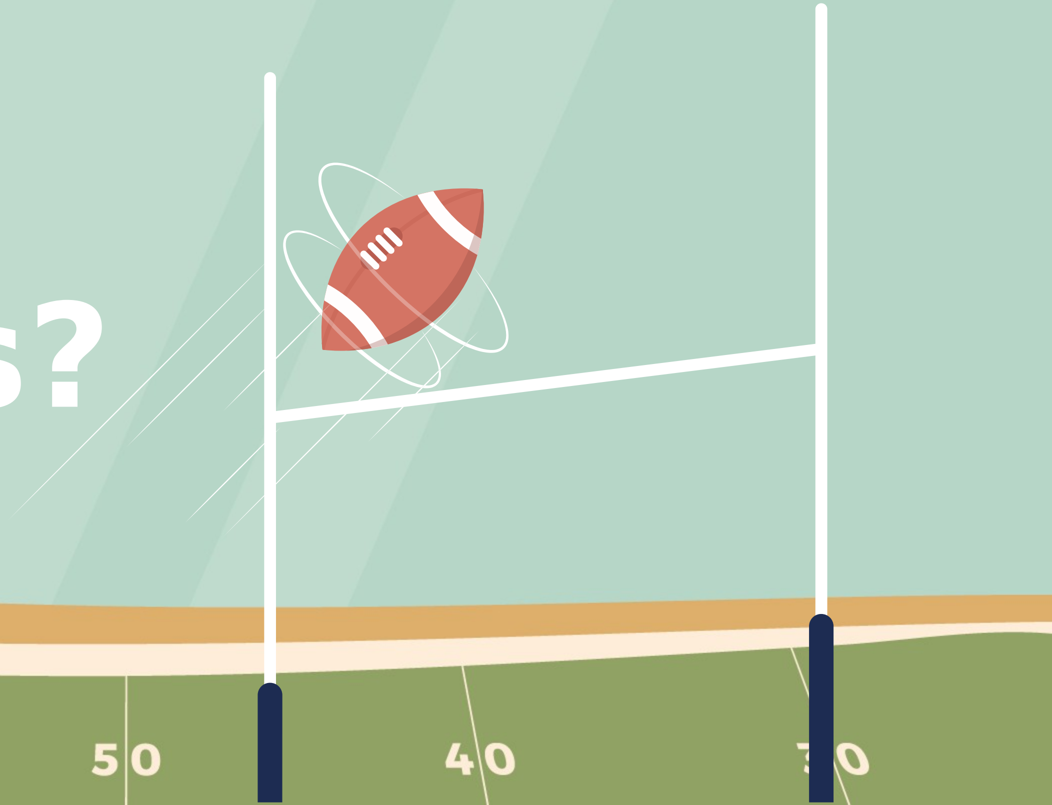
- As expected field goal success decreases as distance increases (Shown in Logistic Curve)
- Teams most often take field goals at the 39.5-49.5 range (29.9%)
- Teams attempt just as many FGS from 30-40 yard to 40-50 yards despite 22.3% lower success rate
- Penalties heavily affect kicks:
  a 5 yard penalty at 45 yards causes an 80% to 72% success drop
- Despite range of kickers these days it may be more statistically viable to go for it on 4th down.

Limitations

Does not account for as we assumed for independence:

- **Weather conditions**
- **Dome vs outdoors**
- **Kicker skill**

Any Questions?



References

1. *2024 NFL Standings & Team Stats*. Pro. (n.d.). https://www.pro-football-reference.com/years/2024/#kicking
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